Kubernative TutedudeAssignment

Kubernetes Deployment Assignment Submission

1. Objective

The objective of this assignment is to deploy a multi-tier application consisting of a **Flask frontend** and an **Express backend** on a local Kubernetes cluster using Minikube.

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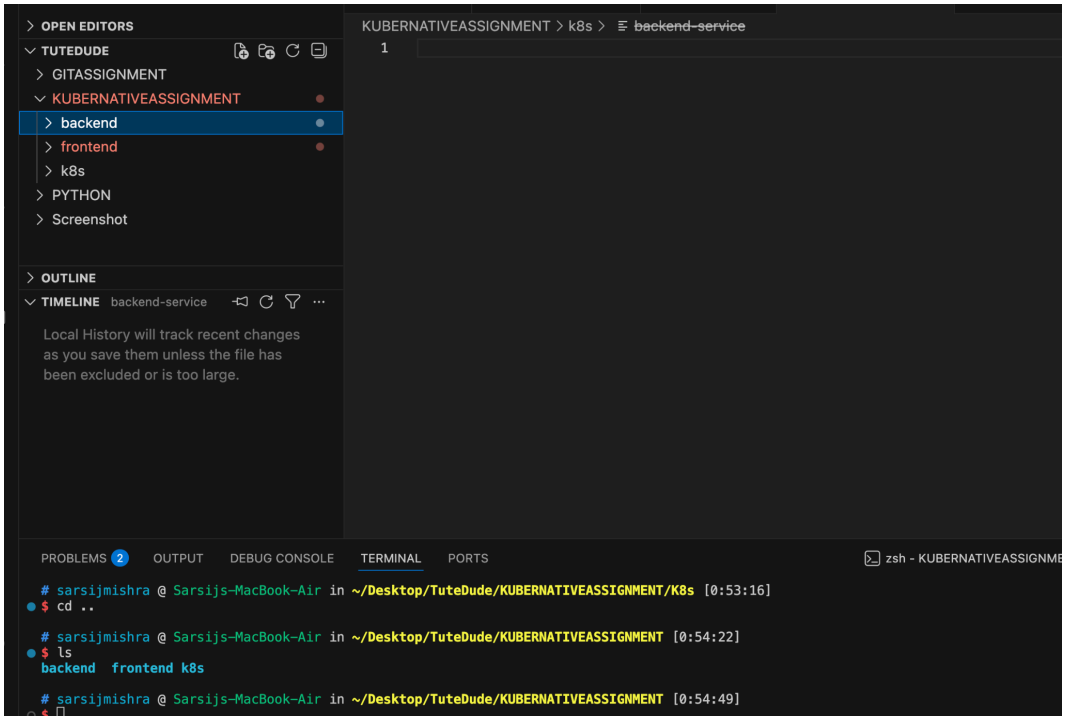
2. Technologies Used

- Flask (Frontend)
- Express.js (Backend)
- Docker
- Kubernetes
- Minikube



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3. Project Architecture

The application consists of two main components:

1. **Flask Frontend**
    - Displays data to the user
    - Sends requests to backend service
2. **Express Backend**
    - Provides API endpoint
    - Returns JSON response

Both components are:

- Containerized using Docker
- Deployed using Kubernetes Deployments
- Exposed using Kubernetes Services

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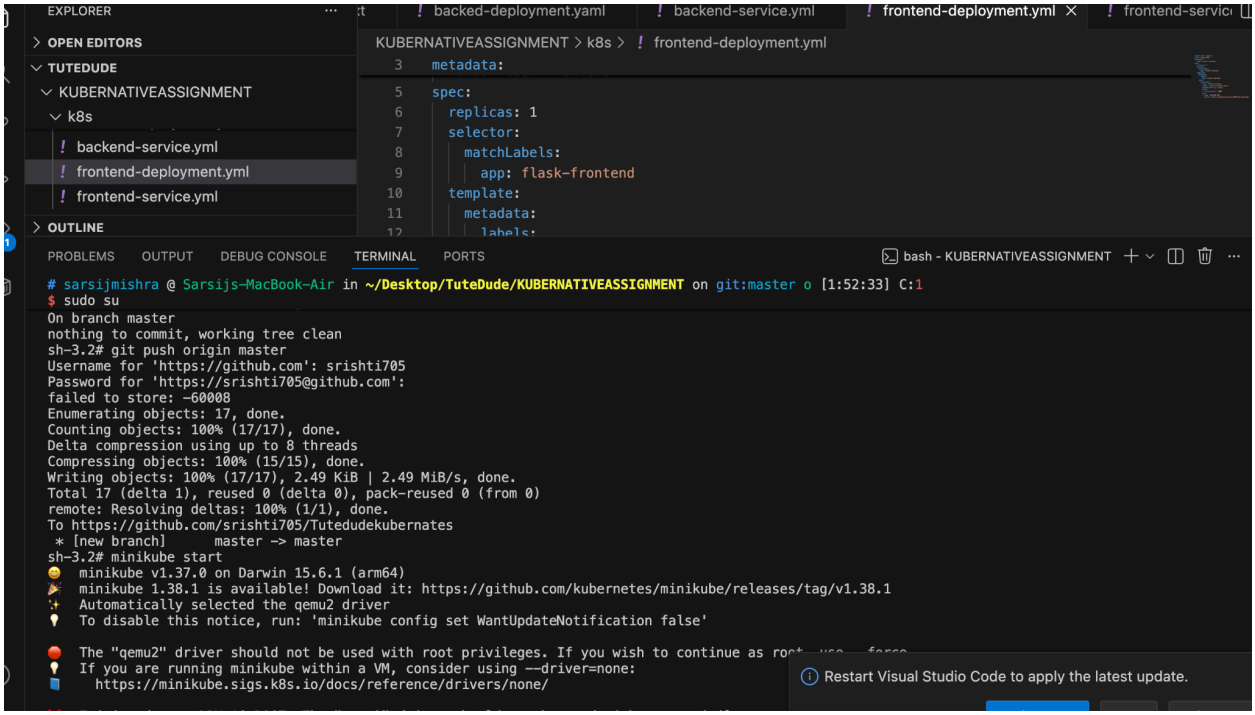
4. Steps Performed

1. Started the Minikube cluster:



minikube start

2.

```
exit

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master o [2:00:14] C:57
$ minikube start
😄  minikube v1.37.0 on Darwin 15.6.1 (arm64)
✨  Automatically selected the qemu2 driver
🌐  Automatically selected the builtin network
❗  You are using the QEMU driver without a dedicated network, which doesn't support `minikube service` & `minikul
To try the dedicated network see: https://minikube.sigs.k8s.io/docs/drivers/qemu/#networking
💾  Downloading VM boot image ...
🎉  minikube 1.38.1 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.38.1
💡  To disable this notice, run: 'minikube config set WantUpdateNotification false'

    > minikube-v1.37.0-arm64.iso....:  65 B / 65 B [---------] 100.00% ? p/s 0s
    > minikube-v1.37.0-arm64.iso:  16.80 MiB / 387.80 MiB  4.33% 1.20 MiB p/s E
```

3. Configured Docker to use Minikube environment:

   eval $(minikube docker-env)

4. Built Docker images:

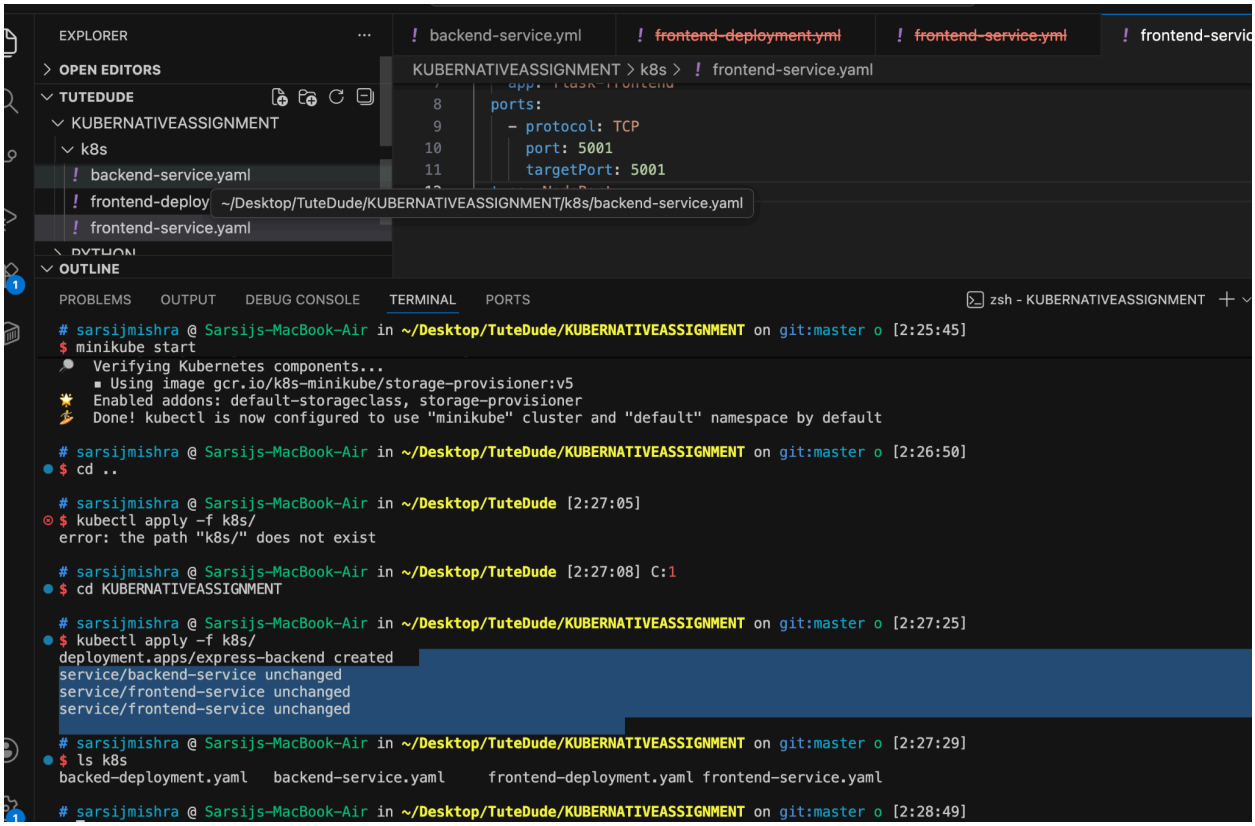
   docker build -t express-backend:latest .
   docker build -t flask-frontend:latest .

5. Created Kubernetes YAML files for:

   - Backend Deployment and Service
   - Frontend Deployment and Service

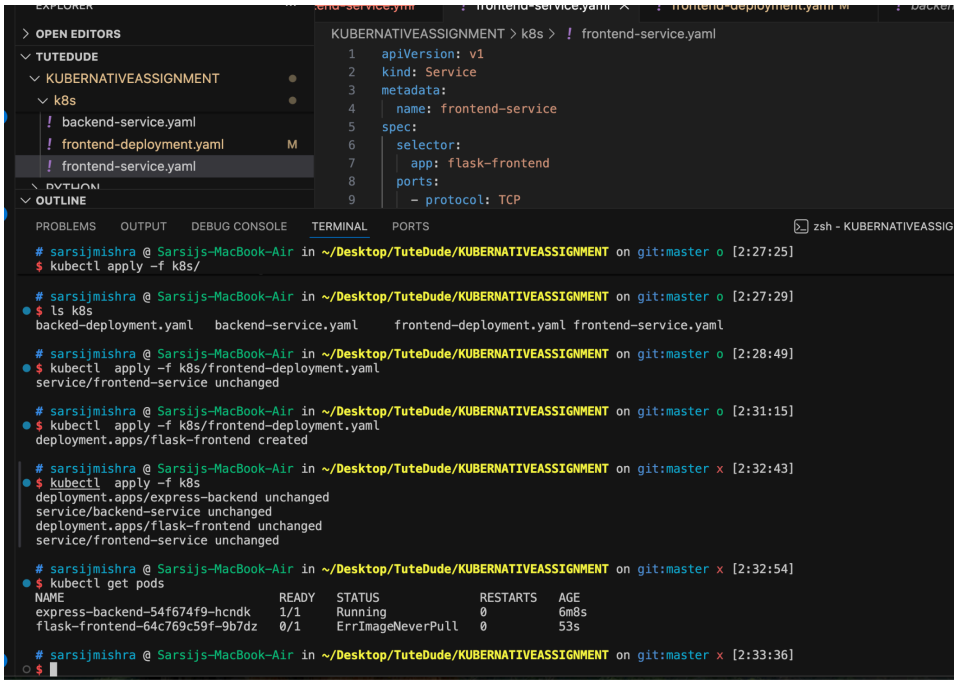
6. Applied Kubernetes configurations:

   kubectl apply -f k8s/backend-deployment.yaml
   kubectl apply -f k8s/backend-service.yaml
   kubectl apply -f k8s/frontend-deployment.yaml
   kubectl apply -f k8s/frontend-service.yaml

7. 

8. Verified resources:

kubectl get pods

9. 

10. kubectl get deployments

```
$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
express-backend-54f674f9-hcndk      1/1     Running             0          6m8s
flask-frontend-64c769c59f-9b7dz     0/1     ErrImageNeverPull   0          53s

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:33:36]
$ kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
express-backend   1/1     1            1           6m59s
flask-frontend    0/1     1            0           104s

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:34:27]
```

11.

```
OPEN EDITORS
TUTEDUDE
KUBERNATIVEASSIGNMENT
k8s
! backend-service.yaml
! frontend-deployment.yaml   M
! frontend-service.yaml
OUTLINE

KUBERNATIVEASSIGNMENT > k8s > ! frontend-service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: frontend-service
5  spec:
6    selector:
7      app: flask-frontend
8    ports:
9      - protocol: TCP

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   zsh - KUBERNATIVEAS

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:36:22]
$ cd frontend
docker build -t flask-frontend:latest .
cd ..
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.144kB
Step 1/7 : FROM python:3.10-slim
 ---> ea297a35336c
Step 2/7 : WORKDIR /app
 ---> Using cache
 ---> ac37036060e9
Step 3/7 : COPY requirements.txt .
COPY failed: file not found in build context or excluded by .dockerignore: stat requirements.txt: file does not exist

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:36:41]
$ kubectl rollout restart deployment flask-frontend
deployment.apps/flask-frontend restarted

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:36:57]
$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
express-backend-54f674f9-hcndk      1/1     Running             0          9m40s
flask-frontend-64c769c59f-9b7dz     0/1     ErrImageNeverPull   0          4m25s
flask-frontend-7d8894ff7b-7hqlw     0/1     ErrImageNeverPull   0          11s

# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:37:09]
$
```

12.

kubectl get services

```
# sarsijmishra @ Sarsijs-MacBook-Air in ~/Desktop/TuteDude/KUBERNATIVEASSIGNMENT on git:master x [2:37:09]
$ kubectl get services
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
backend-service    ClusterIP   10.109.142.110   <none>        5000/TCP         13m
frontend-service   NodePort    10.108.3.131     <none>        5001:31537/TCP   13m
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP          31m
```

13.

14.

15. Accessed application:

    minikube service frontend-service

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5. Kubernetes Resources

- **Backend**
  - Deployment: express-backend
  - Service: ClusterIP
- **Frontend**
  - Deployment: flask-frontend
  - Service: NodePort

- Docker image build process
- Kubernetes deployment commands
- Running pods, deployments, and services
- Final application output in browser

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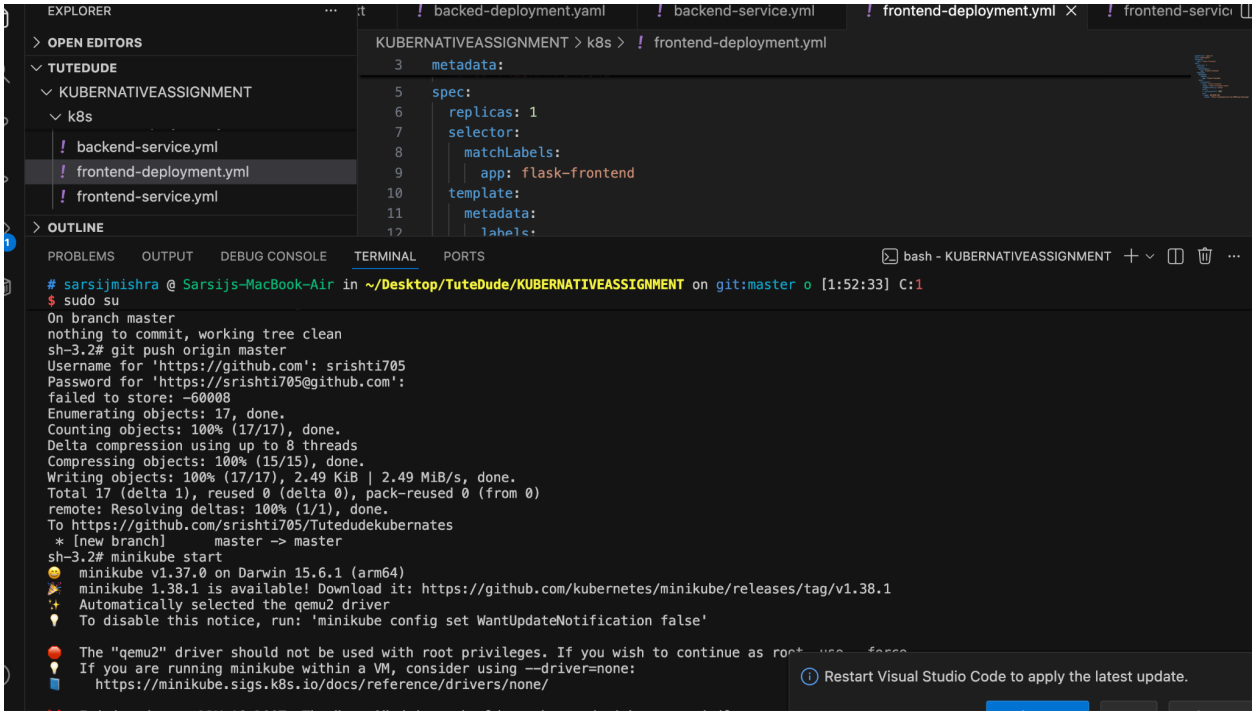
8. GitHub Repository

Repository Link:
(Paste your GitHub repo link here)

Pushes changes to Github :
https://github.com/srishti705/Tutedudekubernates



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9. Conclusion

The application was successfully deployed on a local Kubernetes cluster using Minikube.
This assignment demonstrates:

- Containerization using Docker

- Deployment using Kubernetes
- Service communication within a cluster